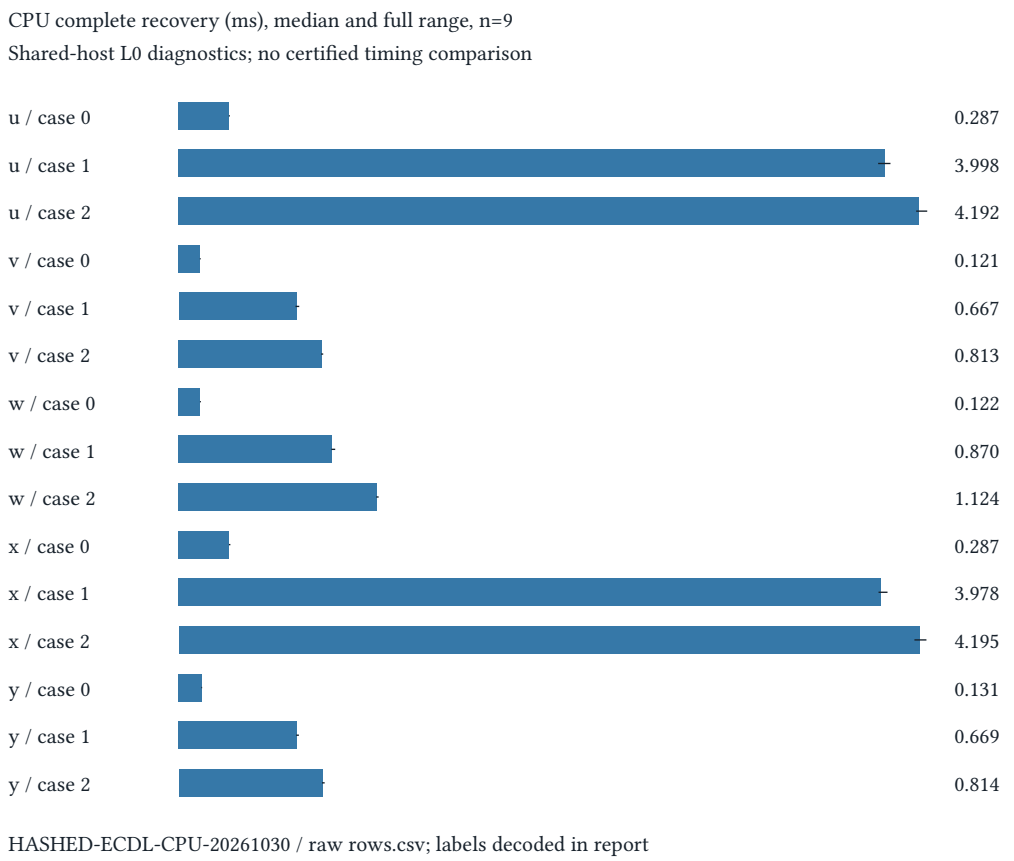
CPU density-aware query scheduling

Shared-host diagnostics only; no certified speedup or hypothesis promotion. Job `hashed-ecdl-cpu-round11-20261030`, October 30, 2026 UTC.

Original baseline u/x: `2bd4588fba60`. Candidate v: `d0ad5a617f59`; batch1 w. Fixed-batch reference y: `abb2795c95cb`. One software thread, AMD EPYC9R45, 2 vCPUs / 4 GiB reserved, shared EKS VM (L0). All earlier controls remain present.

Fixture	Baseline u ms	Fixed y ms	Candidate v ms	y/v
8-bit / 3 slots	0.287191	0.131490	0.120901	1.088
16-bit / 3 slots	3.997944	0.668873	0.667152	1.003
16-bit / 4 slots	4.192356	0.814393	0.813343	1.001

Medians from nine measured repetitions after one warmup. Baseline u/v ratios: 2.375, 5.993, 5.154. The further 10x target remains unmet.



The preselected rule uses progressive batches only when distinct suffix sums cover at least 1/8 of the group. It is a scheduling heuristic, not an early-match guarantee. Prefix order, exact first witnesses and exhaustive negative queries are unchanged. Duplicate tuple sums cannot inflate the unique-key count.

All 150 executions completed (135 measured,15 warmups). Full certificate bytes, scalar checks, replay and corrupted-scalar rejection pass. The complete interval charges setup, discarded trials, elimination, both checks and serialization. Process launch, archive I/O and negative controls are excluded.

Small-fixture ranges: v 0.120111–0.125070 ms, y 0.128741–0.133461 ms. The larger fixtures have overlapping ranges and near-equal medians; neither improvement nor equivalence is certified. Retain this candidate as a diagnostic engineering choice. All 29 release tests and strict Clippy pass.

The export passes file-integrity/completeness checks with five-arm counts and its frozen analyzer identity. Earlier metadata/provenance corrections remain preserved in their own panels. Strict isolation, independent review, matched external/rho comparisons and larger-scale recovery remain unresolved. Protocol: `b8707ca8ac830868131ffa10390cb6273ae1a8d23d50b86e4cea59d2230e1abf`. Raw rows, certificates, build identity and checked export: `round11/results/`.